

STUDENT NUMBER: \_\_\_\_\_

TEACHER'S NAME: \_\_\_\_\_

## BAULKHAM HILLS HIGH SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE  
EXAMINATION

2006

MATHEMATICS  
EXTENSION 1*Time allowed – Two hours  
(Plus five minutes reading time)*

## GENERAL INSTRUCTIONS:

- Attempt ALL questions.
- Start each of the 7 questions on a new page.
- All necessary working should be shown.
- Write your student number at the top of each page of answer sheets.
- At the end of the exam, staple your answers in order behind the cover sheet.

## QUESTION 1

Marks

(a) Find  $\int \frac{dx}{9+16x^2}$ .

2

(b) The line  $y = mx$  makes an angle of  $45^\circ$  with the line  $y = 3x - 4$ . Find the possible values of  $m$ .

3

(c) Find the ratio in which  $P(-15, -10)$  divides the interval  $AB$  where  $A = (3, -1)$ ,  $B = (9, 2)$ .

2

(d) Find  $\lim_{x \rightarrow 0} \frac{\sin 5x}{2x}$ .

1

(e) Use the substitution  $u = x + 1$  to find  $\int_0^8 \frac{2x dx}{\sqrt{x+1}}$ .

4

## QUESTION 2

(a) Find  $\frac{d}{dx} (\sin^{-1} 2x^3)$ .

2

(b) Sketch  $y = 3 \cos^{-1} 2x$ , showing clearly the domain and range.

2

(c) A curve has parametric equations:

$$x = \cos 2\theta$$

$$y = \sin \theta + 1.$$

2

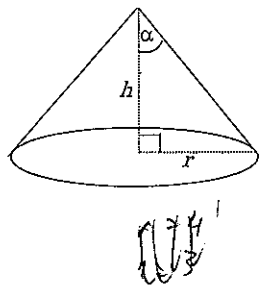
Find its Cartesian equation.

(d) Find  $\int \sin^2 3x dx$ .

2

## QUESTION 2 (Continued)

- (e) Sand is poured at the rate of  $5\text{cm}^3/\text{min}$  into a heap in the shape of a right circular cone whose semi-vertex angle is  $\alpha$  where  $\tan \alpha = \frac{4}{3}$ .



- (i) Show that the volume of the cone of sand is given by:

$$V = \frac{16\pi}{27}h^3.$$

- (ii) Find the rate at which the height is increasing at the instant when the height is 12 cm.

## QUESTION 3

- (a) In the expansion of  $\left(2x + \frac{1}{x^2}\right)^{21}$ , find the term independent of  $x$ .



- (b) When  $P(x)$  is divided by  $x^2 - 4$ , the remainder is  $2x + 3$ . Find the remainder when  $P(x)$  is divided by  $x - 2$ .

- (c) Prove by mathematical induction that  $5^{2n} - 1$  is a multiple of 24 for all integers  $n \geq 1$ .

- (d)  $\alpha, \beta, \gamma$  are the roots of the equation  $x^3 - 2x^2 + kx + 16 = 0$ .

- (i) If two of the roots are equal but opposite in sign, find the value of  $k$ .



- (ii) Find the value of  $\alpha^2 + \beta^2 + \gamma^2$ .

Marks

2

2

3

2

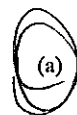
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2

2

## QUESTION 4

Marks



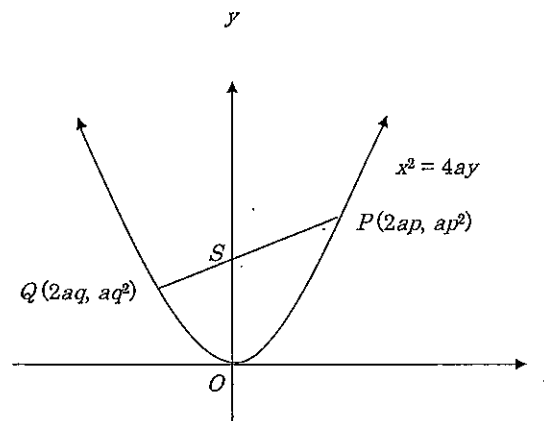
- (a) Find the exact value of  $\tan\left(2\cos^{-1}\frac{7}{25}\right)$ .

3

- (b) The equation  $4\cos\frac{\pi}{2}x - 6 + x = 0$  has a root near 3.5. Use one application of Newton's method to find a second approximation to the root. Give answer to 3 significant figures.

3

- (c)



$P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  are 2 variable points on the parabola  $x^2 = 4ay$ .  $S$  is the focus.



- (i) If  $PQ$  is a focal chord, show that  $pq = -1$ .

2



- (ii)  $R$  is the midpoint of  $PQ$ . Find the coordinates of  $R$  and hence find the equation of the locus of  $R$ .

3



- (iii) Find the length of  $SP$  in terms of  $p$ .

1

## QUESTION 5

(a) Prove the identity  $\frac{2 \tan \theta}{1 + \tan^2 \theta} = \sin 2\theta$ .

(b) Find  $\frac{d}{dx} \left( x \tan^{-1} 2x - \frac{1}{4} \ln(1 + 4x^2) \right)$ .

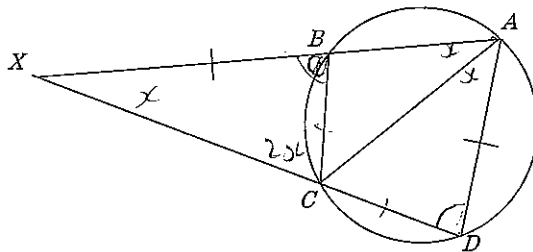
Hence find  $\int_0^{\frac{1}{2}} \tan^{-1} 2x \, dx$  in exact form.

(c) In the figure below, it is given that  $AC$  bisects  $\angle BAD$  and  $BX = AD$ .

Prove that:

(i)  $\triangle BCX \cong \triangle ACD$ .

(ii)  $\triangle ACX$  is isosceles.



Marks

3

2

2

3

2

## QUESTION 6

Marks

(a) A particle moves along the  $x$  axis and its velocity  $v$  m/s at the position  $x$  metres is given by  $v^2 = 30 + 4x - 2x^2$

(i) Prove that the motion is simple harmonic.

2

(ii) Find the centre, and period of the motion.

2

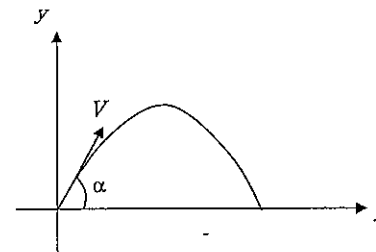
(iii) What is the amplitude of the motion?

2

(iv) Find the maximum speed.

1

(b)



A stone is projected with velocity  $V$  at an angle  $\alpha$ . It hits a target 40m from the point of projection on the ground. On its path, it passes through a point 10 m above the ground and 25 m from the point of projection. [Take  $g = 10\text{m/s}^2$ ]

(i) Given that

$$x = V \cos \alpha t, \quad y = \frac{-gt^2}{2} + V \sin \alpha t.$$

Find the Cartesian equation of the trajectory.

2

(ii) Show that  $\alpha = \tan^{-1} \frac{16}{15}$ .

3

## QUESTION 7

Marks

- (a) (i) Without calculus, sketch the graph of the function.

$$f(x) = x - \frac{1}{x} \quad \text{for } x < 0.$$

1

- (ii) Find the inverse function  $f^{-1}(x)$ .

2

- (iii) Sketch the inverse function  $f^{-1}(x)$  on the same axes in (i).

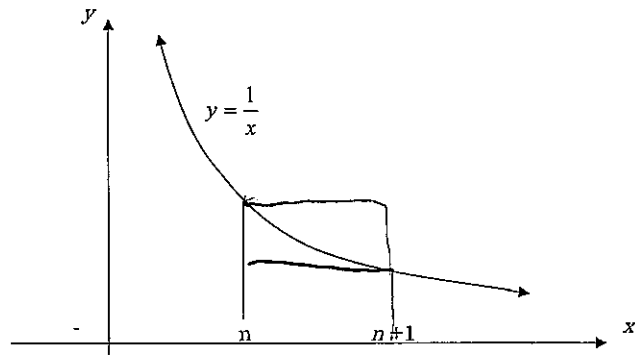
1

- (b) Using the expansion of  $x(1+x)^n$ , show that :

3

$$\binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \dots + (n+1)\binom{n}{n} = (n+2)2^{n-1}.$$

- (c)



- (i) Copy the graph of  $y = \frac{1}{x}$  above and use it to show that:

2

$$\frac{1}{n+1} < \int_n^{n+1} \frac{1}{x} dx < \frac{1}{n}.$$

- (ii) Deduce that  $\left(1 + \frac{1}{n}\right)^n < e < \left(1 + \frac{1}{n}\right)^{n+1}$ .

3

Q1

$$\begin{aligned} \text{a) } \int \frac{dx}{9+16x^2} &= \int \frac{dx}{16(\frac{9}{16}+x^2)} \\ &= \frac{1}{16} \cdot \frac{4}{3} \tan^{-1} \frac{4x}{3} + C \end{aligned}$$

$$\text{b) } \tan 45^\circ = \left| \frac{m-3}{1+3m} \right|$$

$$\begin{aligned} 1 &= \frac{m-3}{1+3m} \quad \text{or} \quad -1 = \frac{m-3}{1+3m} \\ 1+3m &= m-3 \quad -1-3m = m-3 \\ m &= -2 \quad m = \frac{1}{2} \end{aligned}$$

c) Let the ratio be  $m:n$   
 $A(3, -1) \quad B(9, 2)$   
 $m \quad n$

$$(-15, -10) = \left( \frac{3n+9m}{m+n}, \frac{-n+2m}{m+n} \right)$$

$$\begin{aligned} -15 &= \frac{3n+9m}{m+n} \\ -24m &= 18n \quad \therefore \frac{m}{n} = -\frac{3}{4} \end{aligned}$$

$\therefore$  divides externally in the ratio 3:4

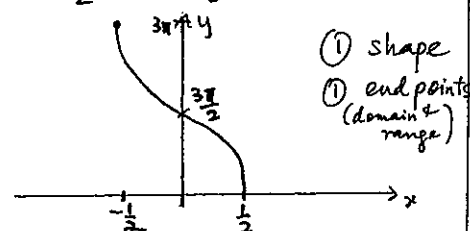
$$\text{d) } \lim_{x \rightarrow 0} \frac{\sin 5x}{2x} = \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \cdot \frac{5}{2} = \frac{5}{2}$$

$$\begin{aligned} \text{e) } \int_0^8 \frac{2x dx}{\sqrt{x+1}} &\quad \text{Let } u = x+1 \\ &\quad du = dx \\ &\quad x=0, u=1 \\ &\quad x=8, u=9 \\ &= \int_1^9 \frac{2(u-1) du}{u^{1/2}} \\ &= \int_1^9 2u^{1/2} - 2u^{-1/2} du \\ &= \left[ 2 \cdot \frac{2}{3} u^{3/2} - 2 \cdot 2u^{1/2} \right]_1^9 \\ &= \left( \frac{4}{3} \cdot 9^{3/2} - 4 \cdot 9^{1/2} \right) - \left( \frac{4}{3} - 4 \right) \\ &= 26\frac{2}{3} \end{aligned}$$

Q2

$$\begin{aligned} \text{a) } \frac{d}{dx} (\sin^{-1} 2x^3) &= \frac{1}{\sqrt{1-4x^6}} \cdot 6x^2 \\ &= \frac{6x^2}{\sqrt{1-4x^6}} \end{aligned}$$

b)  $D: -1 \leq 2x \leq 1 \quad R: 0 \leq y \leq 3\pi$   
 $-\frac{1}{2} \leq x \leq \frac{1}{2}$



$$\begin{aligned} \text{c) } x &= \cos 2\theta \\ y &= \sin \theta + 1 \\ \cos 2\theta &= 1 - 2\sin^2 \theta \\ x &= 1 - 2(y-1)^2 \end{aligned}$$

$$\begin{aligned} \text{d) } \int \sin^2 3x dx &= \frac{1}{2} \int 1 - \cos 6x dx \\ &= \frac{1}{2} \left[ x - \frac{\sin 6x}{6} \right] + C \end{aligned}$$

e) ii)  $\tan \alpha = \frac{r}{h} = \frac{4}{3} \quad \therefore r = \frac{4}{3}h$

$$\begin{aligned} V &= \frac{1}{3} \pi r^2 h \\ &= \frac{1}{3} \pi \left( \frac{16}{9} h^2 \right) \cdot h \\ &= \frac{16}{27} \pi h^3 \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{dV}{dh} &= \frac{16}{9} \pi h^2 \\ \frac{dh}{dt} &= \frac{dV}{dV} \times \frac{dV}{dt} \\ &= \frac{9}{16\pi h^2} \times 5 \text{ cm/min} \\ h=12, \frac{dh}{dt} &= \frac{9}{16\pi 12^2} \times 5 \text{ cm/min} \\ &= \frac{5}{256\pi} \text{ cm/min} \end{aligned}$$

Q3

$$\begin{aligned} \text{a) } (2x + \frac{1}{x^2})^{21} \\ T_{r+1} &= \binom{21}{r} (2x)^{21-r} \cdot \left(\frac{1}{x^2}\right)^r \\ &= \binom{21}{r} 2^{21-r} \cdot x^{21-3r} \end{aligned}$$

For term indep of  $x$ :  $21-3r=0$   
 $r=7$

$$\therefore T_8 = \binom{21}{7} \cdot 2^{14} = 1905131520$$

$$\begin{aligned} \text{b) } P(x) &= (x^2 - 4)Q(x) + 2x + 3 \\ \text{when divided by } x-2, \text{ remainder} &= P(2) \\ \text{remainder} &= (4-4)Q(2) + 4+3 = 7 \end{aligned}$$

c) test  $n=1 \quad 5^2 - 1 = 24$  is a multiple of 24.  
 $\therefore$  true for  $n=1$

Assume true for  $n=k$   
 $5^{2k} - 1 = 24N$

$$\begin{aligned} \text{when } n=k+1 \\ 5^{2(k+1)} - 1 &= 5^{2k+2} - 1 \\ &= 25(5^{2k} + 1) - 1 \\ &= 25(24N+1) - 1 \\ &= 25 \cdot 24N + 24 \\ &= 24(25N+1) \text{ which is a multiple of 24} \end{aligned}$$

$\therefore$  If true for  $n=k$ , it will be true for  $n=k+1$ . Since true for  $n=1$ , it will be true for  $n=2, 3, \dots$

$$\begin{aligned} \text{d) ii) } \text{If } \beta = -\alpha \\ \text{sum of roots} &= \alpha - \alpha + 8 = 8 \\ \text{sub } x=2 &\quad \therefore 8=2 \\ \therefore 2 - 2 \cdot 2^2 + 2k + 16 &= 0 \\ k &= -8 \end{aligned}$$

$$\begin{aligned} \text{ii) } \Sigma \alpha^2 &= (\Sigma \alpha)^2 - 2 \Sigma \alpha \beta \\ &= 2^2 - 2(k) = 20 \end{aligned}$$

Q4

$$\begin{aligned} \text{a) } \tan(2 \cos^{-1} \frac{7}{25}) &= \frac{2 \tan \alpha}{1 - \tan^2 \alpha} \\ &= \frac{2(-\frac{24}{7})}{1 - (-\frac{24}{7})^2} = \frac{336}{527} \end{aligned}$$

$$\begin{aligned} \text{b) } f(x) &= 4 \cos \frac{\pi}{2} x - 6 + x \\ f'(x) &= -4 \cdot \frac{\pi}{2} \sin \frac{\pi}{2} x + 1 \\ x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 3.5 - \frac{(4 \cos \frac{\pi}{2} \cdot 3.5 - 6 + 3.5)}{-2\pi \sin \frac{\pi}{2} (3.5) + 1} \\ &= 3.5 - \frac{0.328427}{5.44288} \\ &= 3.44 \end{aligned}$$

c) i) PQ is focal chord

$$\begin{aligned} m_{ps} &= m_{as} \\ \frac{ap^2 - a}{2ap} &= \frac{aq^2 - a}{2aq} \\ q(p^2 - 1) &= p(q^2 - 1) \\ pq(p-q) + (p-q) &= 0 \\ (p-q)(pq+1) &= 0 \\ p \neq q \quad \therefore pq &= -1 \end{aligned}$$

$$\text{ii) } R = (a(p+q), \frac{a(p^2+q^2)}{2})$$

$$\begin{aligned} \text{Locus is } x &= a(p+q) \\ y &= \frac{a}{2} (p^2+q^2) \end{aligned}$$

$$\begin{aligned} \text{Using } (p+q)^2 &= p^2+q^2+2pq \\ \left(\frac{x}{a}\right)^2 &= \frac{2y}{a} + 2 \\ \text{or } x^2 &= 2a(y+a) \end{aligned}$$

$$\begin{aligned} \text{iii) } SP &= \text{distance of } P \text{ from directrix} \\ &= ap^2 + a \end{aligned}$$

Q5  
(a) LHS =  $\frac{2 \sin \theta}{\sec^2 \theta} \rightarrow \textcircled{1}$   
 $= \frac{2 \sin \theta}{\cos^2 \theta} \times \cos^2 \theta \rightarrow \textcircled{1}$   
 $= 2 \sin \theta \cos \theta$   
 $= \sin 2\theta$   
 $= \text{RHS}$

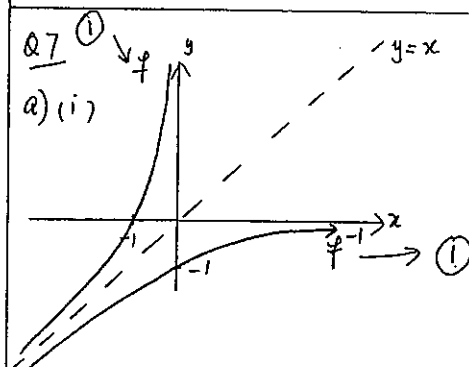
b)  $\frac{d}{dx} (x \tan^{-1} 2x - \frac{1}{4} \ln(1+4x^2))$   
 $= x \cdot \frac{1}{1+4x^2} \cdot 2 + \tan^{-1} 2x \cdot 1 - \frac{1}{4} \cdot \frac{8x}{1+4x^2}$   
 $= \frac{2x}{1+4x^2} + \tan^{-1} 2x - \frac{2x}{1+4x^2}$   
 $= \tan^{-1} 2x$   
 $\therefore \int_0^{\frac{1}{2}} \tan^{-1} 2x \, dx$   
 $= \left[ x \tan^{-1} 2x - \frac{1}{4} \ln(1+4x^2) \right]_0^{\frac{1}{2}}$   
 $= \left( \frac{1}{2} \tan^{-1} 1 - \frac{1}{4} \ln 2 \right) - \left( 0 - \frac{1}{4} \ln 1 \right)$   
 $= \frac{\pi}{8} - \frac{1}{4} \ln 2$   $\textcircled{1}$

(i) To prove  $\triangle BCX \equiv \triangle ACD$   
Proof:  $\angle BAC = \angle DAC$  (AC bisects  $\angle BAD$ )  
 $\therefore BC = DC$  (equal chords subtend equal  $\angle$ s at circum)  
 $\angle XBC = \angle ADC$  (ext  $\angle$  of cyclic quad = int opp.  $\angle$ )  
 $BX = AD$  (given)  
 $\therefore \triangle BCX \equiv \triangle ACD$  (SAS)  
 $\angle CXB = \angle CAD$  (matching  $\angle$ s of congruent  $\Delta$ s)  $\textcircled{1}$   
but  $\angle CAD = \angle CAB$  (given)  
 $\therefore \angle CXB = \angle CAB$   $\textcircled{1}$   
 $\therefore \triangle ACX$  is isosceles (2 equal  $\angle$ s)

Q6  
a)  $v^2 = 30 + 4x - 2x^2$   
ii)  $\ddot{x} = \frac{d}{dx} (\frac{1}{2} v^2)$   
 $= \frac{d}{dx} (15 + 2x - x^2)$   $\textcircled{1}$   
 $= 2 - 2x$   
 $= -2(x-1)$  of the form  $-n^2(x-c)$   $\textcircled{1}$   
 $\therefore$  motion is S.H.  
(ii) centre is  $x=1$   $\textcircled{1}$   
period =  $\frac{2\pi}{n} = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$   $\textcircled{1}$   
(iii) when  $v=0$   $30 + 4x - 2x^2 = 0$   
 $15 + 2x - x^2 = 0$   
 $(3+x)(5-x) = 0$   
 $\therefore x = -3$  or  $5$   $\textcircled{1}$   
amplitude = 4  $\textcircled{1}$   
(iv) max speed when  $\ddot{x} = 0$   
ie at  $x=1$   $\textcircled{1}$   
max speed =  $\sqrt{30 + 4 - 2} = 4\sqrt{2} \text{ m/s}$

(b) (i)  $x = V \cos \alpha t \therefore t = \frac{x}{V \cos \alpha}$   $\textcircled{1}$   
 $y = -\frac{gt^2}{2} + V \sin \alpha t$   
 $= -\frac{g}{2} \left( \frac{x^2}{V^2 \cos^2 \alpha} \right) + V \sin \alpha \cdot \frac{x}{V \cos \alpha}$   
 $y = x \tan \alpha - \frac{gx^2}{2V^2 \cos^2 \alpha}$   $\textcircled{1}$   
(ii)  $x=40, y=0$   
 $\therefore 0 = 40 \tan \alpha - \frac{g(1600)}{2V^2 \cos^2 \alpha}$   $\textcircled{1}$   
 $\frac{g}{V^2 \cos^2 \alpha} = \frac{\tan \alpha}{20}$

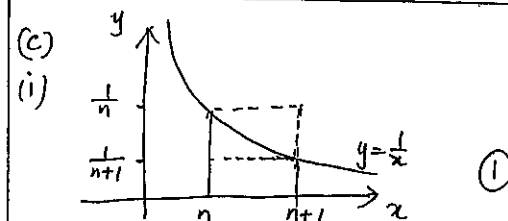
Also  $x=25, y=10$   
 $10 = 25 \tan \alpha - \frac{625g}{2V^2 \cos^2 \alpha}$   $\textcircled{1}$   
 $= 25 \tan \alpha - \frac{625}{2} \cdot \frac{\tan \alpha}{20}$   
 $10 = \frac{75}{8} \tan \alpha \therefore \tan \alpha = \frac{80}{75} = \frac{16}{15}$



(ii) For inverse function,  
let  $x = y - \frac{1}{y}$   
 $y^2 - xy - 1 = 0$   
 $y = \frac{x \pm \sqrt{x^2 + 4}}{2}$   
 $y = \frac{x + \sqrt{x^2 + 4}}{2} \rightarrow \textcircled{1}$   
Since  $R_{f^{-1}}$  is  $y < 0$   
 $f^{-1}(x) = \frac{x - \sqrt{x^2 + 4}}{2} \rightarrow \textcircled{1}$

(b)  $x(1+x)^n = x \left[ \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n \right]$   
 $= \binom{n}{0}x + \binom{n}{1}x^2 + \binom{n}{2}x^3 + \dots + \binom{n}{n}x^{n+1}$   
differentiate both sides  
 $x \cdot n(1+x)^{n-1} + (1+x)^n \cdot 1 = \binom{n}{0} + 2\binom{n}{1}x + 3\binom{n}{2}x^2 + \dots + (n+1)\binom{n}{n}x^n$   
Sub  $x=1$   
 $n \cdot 2^{n-1} + 2^n = \binom{n}{0} + 2\binom{n}{1} + 3\binom{n}{2} + \dots + (n+1)\binom{n}{n}$   
 $2^{n-1}[n+2] = \dots + (n+1)\binom{n}{n}$

- $\textcircled{1}$  differentiate  
 $\textcircled{1}$  Sub  $x=1$   
 $\textcircled{1}$  result



From the graph

area of lower rectangle  $<$  area under curve from  $n$  to  $n+1$   $<$  area of upper rectangle  
ie  $\frac{1}{n+1} < \int_n^{n+1} \frac{1}{x} dx < \frac{1}{n}$

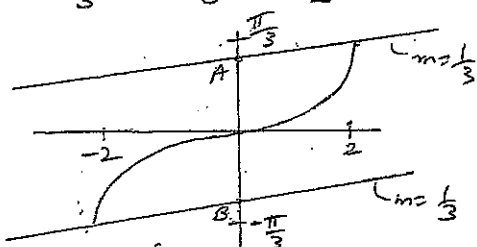
(ii) From  $\int_n^{n+1} \frac{1}{x} dx < \frac{1}{n}$   
 $[\ln x]_n^{n+1} < \frac{1}{n}$   $\textcircled{1}$   
 $\ln \frac{n+1}{n} < \frac{1}{n}$   
 $1 + \frac{1}{n} < e^{\frac{1}{n}}$   
or  $(1 + \frac{1}{n})^n < e$   $\textcircled{1}$   
From  $\int_n^{n+1} \frac{1}{x} dx > \frac{1}{n+1}$   
 $\ln \frac{n+1}{n} > \frac{1}{n+1}$   
 $(1 + \frac{1}{n})^n > e^{\frac{1}{n+1}}$   
 $(1 + \frac{1}{n})^{n+1} > e$   $\textcircled{1}$   
 $(1 + \frac{1}{n})^n < e < (1 + \frac{1}{n})^{n+1}$

6. b)  $y = -10$   
 $y = -10t + c_3 \quad c_3 = V \sin \theta$   
 $y = -10t + V \sin \theta$   
 $y = -5t^2 + V \sin \theta t + c_4$   
 when  $t=0, y=50 \Rightarrow c_4 = 50$   
 $\therefore y = -5t^2 + V \sin \theta t + 50$   
 (ii) when  $t=5, x=100$  and  $y=0$   
 $\therefore 100 = 5V \cos \theta$   
 $V \cos \theta = 20 \quad \text{--- (1)}$   
 $0 = -125 + 5V \sin \theta + 50$   
 $75 = 5V \sin \theta$   
 $15 = V \sin \theta \quad \text{--- (2)}$   
 $\frac{(2)}{(1)} \Rightarrow \frac{3}{4} = \tan \theta$   
 $\theta = 36^\circ 52'$   
 $\therefore V \cos 36^\circ 52' = 20$   
 $V = \frac{20}{\cos 36^\circ 52'}$   
 $= 25 \text{ m/s.}$

(iii) At impact  $V = \sqrt{(u)^2 + (v)^2}$   
 when  $t=5, V=25$   
 $x = 25 \cos 36^\circ 52'$   
 $= 20 \text{ m/s.}$   
 $y = -10(5) + 25 \sin 36^\circ 52'$   
 $= 25 \text{ m/s.}$

$V = \sqrt{(20)^2 + (25)^2}$   
 $= \sqrt{25} \frac{\text{m/s}}{(40.3)} \quad \text{--- (3)}$   
 7a) (i)  $P(\text{Winning}) = \frac{3}{6} \times \frac{3}{6} = \frac{1}{4}$   
 $P(\text{Losing}) = P(B) + P(\text{other black})$   
 $= \frac{1}{6} + \frac{5}{6} \times \frac{1}{6}$   
 $= \frac{11}{36}$   
 (ii)  $P(\text{Win first go}) = \frac{1}{4}$   
 $P(\text{Win 2nd}) = P(\text{Draw then win})$   
 $= \left(\frac{4}{9}\right) \times \frac{1}{4}$   
 $P(\text{Win 3rd}) = P(\text{Draw} \rightarrow \text{Draw} \rightarrow \text{Win})$   
 $= \frac{4}{9} \times \frac{4}{9} \times \frac{1}{4}$   
 $= \left(\frac{4}{9}\right)^2 \times \frac{1}{4}$   
 $\therefore P(\text{Win 1st, 2nd, 3rd...}) =$   
 $\frac{1}{4} + \frac{1}{4} \times \frac{4}{9} + \frac{1}{4} \times \left(\frac{4}{9}\right)^2 = 1$   
 (iii)  $P(\text{Wins}) =$   
 $\frac{1}{4} \left(1 + \frac{4}{9} + \left(\frac{4}{9}\right)^2 + \left(\frac{4}{9}\right)^3 + \dots\right)$   
 $= \frac{1}{4} \left(\frac{1}{1 - \frac{4}{9}}\right)$   
 $= \frac{1}{4} \times \frac{9}{5} = \frac{9}{20} \quad \text{--- (6)}$

7b)  $y = a \sin^{-1} bx$   
 range  $-\frac{\pi}{3} \leftrightarrow \frac{\pi}{3}$   
 Now  $y = \sin^{-1} x$   
 range  $-\frac{\pi}{2} \leftrightarrow \frac{\pi}{2}$   
 $\frac{\pi}{3}$  is  $\frac{2}{3}$  of  $\frac{\pi}{2} \therefore a = \frac{2}{3}$   
 $y = \sin^{-1} bx$   
 $\frac{y}{a} = \sin^{-1} \frac{bx}{a}$   
 $x = \frac{1}{b} \sin \frac{y}{a}$   
 range is  $-2 \leftrightarrow 2$  (twice that of  $y = \sin^{-1} x$ )  
 $\therefore \frac{1}{b} = 2$   
 $b = \frac{1}{2}$   
 $a = \frac{2}{3} \quad b = \frac{1}{2}$   
 (iii)  $y = \frac{2}{3} \sin^{-1} \frac{x}{2}$   
 $y' = \frac{2}{3} \cdot \frac{1}{\sqrt{1 - \left(\frac{x}{2}\right)^2}}$   
 $= \frac{2}{3} \cdot \frac{1}{\sqrt{4 - x^2}}$   
 $= \frac{2}{3} \cdot \frac{1}{\sqrt{4 - x^2}}$   
 when  $x=0, y' = \frac{1}{3}$

(iii)  $\frac{x}{3} + c = \frac{2}{3} \sin^{-1} \frac{x}{2}$   
  
 $\therefore$  Require to solve  $y = \frac{x}{3} + c$   
 $y = \frac{2}{3} \sin^{-1} \frac{x}{2}$  simultaneously  
 $C$  will lie between  $A$  and  $B$ .  
 At  $A$   
 $y = \frac{x}{3} + A$  passes through  
 $(2, \frac{\pi}{3})$   
 $\therefore \frac{\pi}{3} = \frac{2}{3} + A$   
 $\therefore A = \frac{\pi}{3} - \frac{2}{3}$   
 At  $B, y = \frac{x}{3} + B$  passes through  
 $(-2, -\frac{\pi}{3})$   
 $-\frac{\pi}{3} = \frac{-2}{3} + B$   
 $\therefore B = \frac{2 - \pi}{3}$   
 $\therefore \frac{2 - \pi}{3} \leq c \leq \frac{\pi - 2}{3}$

1 mark for recognising  
 need to solve  $y = \frac{x}{3} + c$   
 $y = \frac{2}{3} \sin^{-1} \frac{x}{2}$  simultaneously

(6)